

Data Science Project Preparation – Requirements

1. Project Idea

- Think of a **real-world problem or question** you are curious about.
- Your project must have a **clear objective** (e.g., "predict house prices," "analyze student study habits," "classify types of music," "detect fake news").
- The idea should be **specific and feasible**—not too broad, but still meaningful.

📄 **Deliverable:** Write a short description (3–5 sentences) explaining:

- The problem/question you want to solve.
- Why this topic is interesting or important.
- What type of data you think you will need.

2. Data Collection

You are responsible for **finding or building your dataset**. Here are the methods you can use:

1. **Search online** for open datasets (Kaggle, UCI Machine Learning Repository, government open data portals, etc.).
2. **Download structured data** (CSV, JSON, Excel) that is publicly available.
3. **Collect your own data** (e.g., surveys, web scraping, personal observations).

⚠️ **Rules:**

- Dataset must have at least **200 rows (records)** and at least **5 features (columns)**.
- Data must be **relevant** to your project idea.
- Do not use datasets that are too “overused” (like the Titanic dataset) unless you add your own twist.
- If you create your own dataset, document how you collected it.

📄 **Deliverable:** Provide the raw dataset file (CSV/Excel/JSON) and the link/source where you found it (if applicable).

3. Data Documentation

For every dataset, you must explain:

1. **Source** – where you got the data.
2. **Size** – number of rows and columns.
3. **Features** – list of the columns with a short description of each.
4. **Potential Issues** – any missing values, errors, or biases you expect.

📄 **Deliverable:** A short report (1–2 pages, or a Google Doc) describing your dataset.

4. Submission Requirements

- One project idea description (short paragraph).
 - One dataset file (CSV, Excel, or JSON).
 - One short dataset documentation report.
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5. Evaluation Criteria

You will be graded on:

- **Clarity of project idea** (Is it meaningful and specific?)
 - **Quality of dataset** (Is it relevant, large enough, and useful?)
 - **Effort in data collection** (Did you go beyond just downloading the first dataset you found?)
 - **Documentation** (Did you clearly explain the dataset and its features?)
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☒ **Deadline:** [10/31/2025] ☒ **Submission method:** [Google Classroom]